

Red Hat OpenShift Data Foundation 4.22

Deploying OpenShift Data Foundation on two-node clusters

Instructions on deploying OpenShift Data Foundation on two-node clusters with DRBD

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Abstract

Read this document for instructions about how to install Red Hat OpenShift Data Foundation on two-node OpenShift clusters using Distributed Replicated Block Device (DRBD) for high availability.

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PREFACE

This document describes how to deploy Red Hat OpenShift Data Foundation on two-node clusters.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

PREFACE

Red Hat OpenShift Data Foundation supports deployment on two-node Red Hat OpenShift Container Platform (RHOCP) bare metal clusters with fencing and Distributed Replicated Block Device (DRBD) enabled. This architecture is suitable for distributed and edge deployments where the goal is to minimize hardware footprint while maintaining High Availability (HA).



NOTE

Two-node cluster deployments with DRBD are supported only on bare metal infrastructure. See the Planning your deployment guide and the Preparing to deploy two-node cluster chapter for more information about deployment requirements.

To deploy OpenShift Data Foundation on a two-node cluster, follow the deployment process in the Deploy using local storage devices with DRBD section.

Additional resources

- Planning your deployment
- Preparing to deploy two-node cluster

CHAPTER 1. PREPARING TO DEPLOY TWO-NODE CLUSTER

Before installing OpenShift Data Foundation on a two-node cluster, you must create persistent volumes manually for the OSDs on both nodes.

Prerequisites

- You have a two-node OpenShift cluster deployed on bare metal infrastructure.
  Verify the cluster topology:

  ```
  oc get infrastructure cluster -o jsonpath='{.status.controlPlaneTopology}'
  ```

  The output should be **DualReplica**.

- You have identified the storage devices on both nodes that will be used for OSDs.
- Storage devices must be SSD type with minimum size of 500 GB.
- Storage device sizes must match across both nodes.
- For the list of packages to be mirrored for installing OpenShift Data Foundation in disconnected environments, see the Disconnected environment section in the *Planning your deployment* guide.

Procedure

1. Create the storage class:

   ```
   oc create -f lso-storageclass.yml
   ```

   Example lso-storageclass.yml

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: localblock
provisioner: kubernetes.io/no-provisioner
volumeBindingMode: WaitForFirstConsumer
```

1. Create the persistent volumes:
   Edit the **pv.yaml** file to match your environment:

   - Device path by disk-by-id (for example, **/dev/disk/by-id/wwn-0x58ce38ee2311e971**)
   - Storage size (for example, **storage: 100Gi**)
   - Node affinity (for example, **values: - master-1**)

     ```
     oc create -f pv.yml
     ```

     Example pv.yml

```
apiVersion: v1
kind: PersistentVolume
```

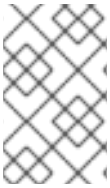
```
metadata:
  name: local-pv-1
spec:
  capacity:
    storage: 500Gi
  accessModes:
  - ReadWriteOnce
  persistentVolumeReclaimPolicy: Delete
  storageClassName: localblock
  local:
    path: /dev/disk/by-id/wwn-0x58ce38ee2311e971
  nodeAffinity:
    required:
      nodeSelectorTerms:
      - matchExpressions:
        - key: kubernetes.io/hostname
          operator: In
          values:
          - master-0
```

+ Repeat this step to create PVs for all OSD disks on both nodes.

Verification

- Verify the creation of PVs:

```
oc get pv
```



NOTE

If you are running OpenShift Container Platform with RHEL KVM, you must assign a serial number to your VM disk. Otherwise, the VM disk cannot be identified after reboot. You can use the **virsh edit <VM>** command to add the **<serial>mydisk</serial>** definition.

CHAPTER 2. DEPLOY OPENSHIFT DATA FOUNDATION ON TWO-NODE CLUSTER

You can deploy OpenShift Data Foundation on a two-node bare metal cluster where OpenShift Container Platform is already installed. When you deploy OpenShift Data Foundation on a two-node OpenShift Container Platform cluster using local storage devices and Distributed Replicated Block Device (DRBD), you create internal cluster resources with high availability. This approach uses DRBD for the floating monitor to ensure data consistency across both nodes.

Perform the following steps to deploy OpenShift Data Foundation:

1. Install the Red Hat OpenShift Data Foundation Operator .
2. Configure DRBD for two-node cluster.
3. Create an OpenShift Data Foundation cluster on two-node cluster .

2.1. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see the *How to use dedicated worker nodes for Red Hat OpenShift Data Foundation* section in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.

4. Click **Install**.
5. Set the following options on the **Install Operator** page:
    a. Update Channel as **stable-4.22**.
    b. Installation Mode as **A specific namespace on the cluster**.
    c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
    d. Select Approval Strategy as **Automatic** or **Manual**.
    If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

    If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
    e. Ensure that the **Enable** option is selected for the **Console plugin**.
    f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
    - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
    - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

2.2. CONFIGURING DRBD FOR TWO-NODE CLUSTER

Configure DRBD (Distributed Replicated Block Device) for the floating monitor after installing the OpenShift Data Foundation operator.

Prerequisites

- You have installed the OpenShift Data Foundation operator.
- You have a dedicated disk or partition on both nodes for the floating monitor with minimum size of 10 GB (recommended less than 50 GB).
- The disk must be SSD type and the same size on both nodes.
- You have **oc** CLI access to the cluster.

Procedure

1. Fetch the DRBD setup script from the configmap:

```
oc get configmap rook-ceph-drbd-setup-script -n openshift-storage -o jsonpath='{.data.script}'
| base64 -d > drbd-setup && chmod +x drbd-setup
```



NOTE

The script must be run from a shell that has **oc** access to the cluster.

2. View the help text to understand available options:

```
./drbd-setup -h
```

Example output

```
Usage:
  ./drbd-setup -d <path>
  ./drbd-setup -d0 <path0> -d1 <path1>
  ./drbd-setup -l

Backing paths are raw block device paths (e.g. /dev/sdb). Use the PATH column from -l.
Disks must be SSD-class (ROTA 0) and same size on both nodes.

  -d PATH
     One path used on both nodes. Choose this when each machine has the replica disk at the
     same device name (both nodes use e.g. /dev/sdb for the DRBD lower layer).

Use -d0/-d1 when the two nodes use different paths; do not combine -d with -d0/-d1.

  -d0 PATH   Path on node 0 only (first node name after sorting all cluster nodes).
  -d1 PATH   Path on node 1 only (second node).

Discovery:
  -l    List block devices on each node (NAME, PATH, SIZE, ROTA, TYPE, FSTYPE).

DRBD options:
  --drbd-conf-path PATH  Host path to drbd.conf (default /etc/drbd.conf)
  --drbd-dir-path PATH   Host dir for resource snippets (default /etc/drbd.d)
  --drbd-resource NAME   Logical DRBD resource name in config (default r0)
  --drbd-device PATH     DRBD upper device node path, same on both nodes (default /dev/drbd0)
  --drbd-port N          TCP port for DRBD replication (default 7794)

General:
  -h    Show this help and exit

Environment:
  Defaults are documented on each assignment near the top of this script.
  OUTPUT_CM_NS / OUTPUT_CM_NAME — namespace and name of the summary ConfigMap
(floating mon).
```

1. List available disks on both nodes:

```
./drbd-setup -l
```

2. Run the setup script specifying the disk path for the floating monitor:

```
./drbd-setup -d /dev/sdb
```



NOTE

- If the script fails intermittently at any step, re-run the script.
- Use **-d** when both nodes use the same device path.
- Use **-d0** and **-d1** when nodes use different device paths.

Verification

- Verify the DRBD configuration ConfigMap:
  A ConfigMap named **drbd-configure** is created in the **openshift-storage** namespace:

```
oc get cm drbd-configure -n openshift-storage -o yaml
```

Example output

```
apiVersion: v1
data:
  BLOCK_DEVICE_PATH_NODE_0: /dev/sdb
  BLOCK_DEVICE_PATH_NODE_1: /dev/sdb
  DISK_BY_ID_NODE_0: /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_drive-scsi1-0-0-0
  DISK_BY_ID_NODE_1: /dev/disk/by-id/scsi-0QEMU_QEMU_HARDDISK_drive-scsi1-0-0-0
  DRBD_CONF_PATH: /etc/drbd.conf
  DRBD_DEVICE_NAME: /dev/drbd0
  DRBD_DIR_PATH: /etc/drbd.d
  DRBD_PORT: "7794"
  DRBD_RESOURCE_NAME: r0
  DRBD_UTILS_IMAGE: registry.redhat.io/odf4/odf-drbd-
rhel9@sha256:af2fc67f7690944008fe56bea65adf2bfa18d7fc437e2b2c21474de3520e3e90
  DRBD_VERSION: 9.2.18
  NODE_0_IP: 192.168.111.20
  NODE_0_NAME: master-0
  NODE_1_IP: 192.168.111.21
  NODE_1_NAME: master-1
  SETUP_TIMESTAMP: "2026-05-18T13:01:44Z"
kind: ConfigMap
metadata:
  name: drbd-configure
  namespace: openshift-storage
```

- Verify the **drbd-autostart** DaemonSet is present in the **openshift-kmm** namespace:

```
oc get daemonset drbd-autostart -n openshift-kmm
```

- Check DRBD status on both nodes:

```
oc debug -q node/master-0 -- chroot /host podman run --rm --privileged \
  -v /dev:/dev -v /etc/drbd.conf:/etc/drbd.conf -v /etc/drbd.d:/etc/drbd.d \
  --hostname master-0 --net host registry.redhat.io/odf4/odf-drbd-
rhel9@sha256:af2fc67f7690944008fe56bea65adf2bfa18d7fc437e2b2c21474de3520e3e90 \
  drbdadm -c /etc/drbd.conf status r0
```

Repeat the command for **master-1**.

2.3. CREATING OPENSHIFT DATA FOUNDATION CLUSTER ON TWO-NODE CLUSTER

Create a OpenShift Data Foundation storage cluster on a two-node deployment after completing DRBD configuration, using the OpenShift Web Console to provision the storage system within the constraints of the two-node architecture.

Prerequisites

- You have completed the steps in Preparing to deploy two-node cluster.
- You have installed the OpenShift Data Foundation operator.
- You have completed the steps in Configuring DRBD for two-node cluster.
- The **drbd-configure** ConfigMap exists in the **openshift-storage** namespace.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation → Storage Systems → Create StorageSystem**.
2. In the **Backing storage** page, select the storage class created earlier and click **Next**.
3. Follow the wizard to configure the storage cluster according to your requirements.
4. Wait for the StorageCluster status to become **Ready**.

Verification

- Verify the StorageCluster status:

```
oc get storagecluster -n openshift-storage
```

  The status should be **Ready**.

- Verify all pods are running:

```
oc get pods -n openshift-storage
```

- Verify the floating monitor is deployed correctly:

```
oc get pods -n openshift-storage | grep rook-ceph-mon
```

- Check Ceph cluster health:

```
oc rsh -n openshift-storage $(oc get pods -n openshift-storage -l app=rook-ceph-tools -o name) ceph status
```



NOTE

- After node recovery, if Ceph status shows OSD pod down, try to restart the OSD pods to get them back.
- Two-node clusters support applications using RBD (RADOS Block Device) and CephFS storage with both ReadWriteOnce (RWO) and ReadWriteMany (RWX) access modes.

2.4. TROUBLESHOOTING TWO-NODE DEPLOYMENTS

During node failover events, the floating monitor might enter an unexpected **drbd-initfailover** state if the incorrect node is marked as DRBD secondary.

Workaround

You can manually set both nodes to DRBD Secondary to reset the DRBD role state:

```
oc debug node/"$Node" -- chroot /host sudo podman run --rm --privileged \
  -v /dev:/dev -v /etc/drbd.conf:/etc/drbd.conf -v /etc/drbd.d:/etc/drbd.d \
  --hostname "$Node" registry.redhat.io/odf4/odf-drbd-
rhel9@sha256:af2fc67f7690944008fe56bea65adf2bfa18d7fc437e2b2c21474de3520e3e90 \
  drbdadm secondary r0
```

Replace **$Node** with the node name (for example, **master-0** or **master-1**). Run this command for both nodes.

2.5. FEATURES NOT SUPPORTED IN TWO-NODE DEPLOYMENTS

Several OpenShift Data Foundation features are not supported in the two-node deployment.

- Multicloud Object Gateway (NooBaa)
- Network File System (NFS)
- RADOS Object Gateway (RGW)
- Disaster Recovery (Regional DR and Metro-DR)
- PodDisruptionBudgets (PDBs)
- Monitor failover
- Multus networking
- External PostgreSQL
- Node selection
- Performance/resource profiles
- Automatic capacity scaling
- Host networking
- Erasure Coded (EC) pools

- External clients (HCI/Provider mode)